./tools/run_validation_workflow.sh

==================================================

Bloodhound Validation Workflow

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Step 1: Running Lambda smoke test...

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Bloodhound Lambda Smoke Test

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Checking Lambda existence...

✓ Lambda function exists

Fetching Lambda environment variables...

✓ Retrieved Lambda environment variables

Validating critical environment variables...

Local mode detected — validating against .env

✓ Environment variable validation passed

Deployed Lambda environment variables:

{
  "SLACK_DESTROY_CONFIRM_TOKEN": "CONFIRM",
  "KEEP_TAG_VALUE": "true",
  "SLACK_ALLOWED_USER_IDS": "",
  "TEARDOWN_ALLOW_ALL": "false",
  "SLACK_ALERT_CHANNEL_ID": "C0A4YLV0HNY",
  "SLACK_SCAN_CHANNEL_ID": "C0A4YLV0HNY",
  "SLACK_ENABLED": "true",
  "APPLY_CHANGES": "false",
  "REGION_MODE": "explicit",
  "RDS_FINAL_SNAPSHOT": "false",
  "SLACK_SIGNING_SECRET": "a77fd7b559aaf8c17eb683ab493741f4",
  "BUDGET_OVER_DAYS": "2",
  "TEARDOWN_SIMULATE": "true",
  "COHORT_TOTAL_BUDGET_USD": "3000",
  "REGIONS": "us-east-1,us-east-2,us-west-1,us-west-2",
  "TEARDOWN_TARGET_IDS": "",
  "KEEP_TAG_KEY": "bloodhound:keep",

"COHORT_LENGTH_MONTHS": "7",

"COHORT_START_YYYY_MM": "2025-12",

"SLACK_ALLOWED_CHANNEL_IDS": "",

"SLACK_BOT_TOKEN": "xoxb-14576256133-10645391136832-23tkIVIh3kgwgfz4rgDR9W5L"

}

✓ Environment variables retrieved

Checking CloudWatch log group...

✓ Log group exists

Checking Lambda Function URL...

✓ Function URL configured

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Smoke test completed successfully

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Smoke test passed.

Step 2: Starting controlled teardown validation...

Validation Run ID: 20260326_205446

Validation Run ID: 20260326_205446

Optional: stream Lambda logs during validation.

Open a second terminal and run:

aws logs tail /aws/lambda/BloodhoundLambdaV2 --region us-west-2 --follow

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Bloodhound Controlled Teardown Validation

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Step 1: Creating disposable EC2 instance

Initializing the backend...

Initializing provider plugins...

- terraform.io/builtin/terraform is built in to Terraform

- Reusing previous version of hashicorp/archive from the dependency lock file

- Reusing previous version of hashicorp/aws from the dependency lock file

- Using previously-installed hashicorp/archive v2.7.1

- Using previously-installed hashicorp/aws v6.27.0

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see

any changes that are required for your infrastructure. All Terraform commands

should now work.

If you ever set or change modules or backend configuration for Terraform,

rerun this command to reinitialize your working directory. If you forget, other

commands will detect it and remind you to do so if necessary.

terraform_data.build_lambda_pkg: Refreshing state... [id=5ad4c1d0-1e07-ae07-cca2-ea66b293a381]

data.archive_file.lambda_zip: Reading...

data.archive_file.lambda_zip: Read complete after 0s [id=b1a59953cc20d5497c31342a65ca9eb3aaa66168]

data.aws_iam_policy_document.lambda_assume: Reading...

data.aws_iam_policy_document.bloodhound: Reading...

data.aws_ami.amazon_linux_2: Reading...

data.aws_subnets.default: Reading...

data.aws_caller_identity.current: Reading...

data.aws_iam_policy_document.lambda_assume: Read complete after 0s [id=2690255455]

data.aws_iam_policy_document.bloodhound: Read complete after 0s [id=3536511632]

aws_iam_role.lambda_role: Refreshing state... [id=bloodhound-v2-role]

aws_iam_policy.bloodhound_policy: Refreshing state... [id=arn:aws:iam::388691194728:policy/bloodhound-v2-policy]

data.aws_caller_identity.current: Read complete after 0s [id=388691194728]

terraform_data.account_guard: Refreshing state... [id=21ae472e-35bd-81b6-76c9-6a7f8b2cc399]

aws_iam_role_policy_attachment.bloodhound_attach: Refreshing state... [id=bloodhound-v2-role/arn:aws:iam::388691194728:policy/bloodhound-v2-policy] aws_iam_role_policy_attachment.basic_logs: Refreshing state... [id=bloodhound-v2-role/arn:aws:iam::aws:policy/service-role/AWSLambdaBasicExecutionRole]

aws_lambda_function.bloodhound_v2: Refreshing state... [id=BloodhoundLambdaV2]

data.aws_subnets.default: Read complete after 1s [id=us-west-2]

data.aws_ami.amazon_linux_2: Read complete after 1s [id=ami-0534a0fd33c655746]

aws_lambda_permission.function_url_invoke: Refreshing state... [id=AllowFunctionInvoke]

aws_lambda_permission.function_url_public: Refreshing state... [id=AllowFunctionUrlPublic]

aws_lambda_alias.bloodhound_prod: Refreshing state... [id=arn:aws:lambda:us-west-2:388691194728:function:BloodhoundLambdaV2:prod]

aws_lambda_function_url.bloodhound_url: Refreshing state... [id=BloodhoundLambdaV2]

Terraform used the selected providers to generate the following execution

plan. Resource actions are indicated with the following symbols:

+ create

Terraform will perform the following actions:

```
# aws_instance.bloodhound_teardown_test[0] will be created
+ resource "aws_instance" "bloodhound_teardown_test" {
    + ami                                  = "ami-0534a0fd33c655746"
    + arn                                  = (known after apply)
    + associate_public_ip_address          = (known after apply)
    + availability_zone                    = (known after apply)
    + disable_api_stop                     = (known after apply)
    + disable_api_termination              = (known after apply)
    + ebs_optimized                        = (known after apply)
    + enable_primary_ipv6                  = (known after apply)
    + force_destroy                        = false
    + get_password_data                    = false
    + host_id                              = (known after apply)
    + host_resource_group_arn              = (known after apply)
    + iam_instance_profile                 = (known after apply)
    + id                                   = (known after apply)
    + instance_initiated_shutdown_behavior = (known after apply)
    + instance_lifecycle                   = (known after apply)
    + instance_state                       = (known after apply)
    + instance_type                        = "t3.micro"
```

```
+ ipv6_address_count                 = (known after apply)
+ ipv6_addresses                     = (known after apply)
+ key_name                           = (known after apply)
+ monitoring                         = (known after apply)
+ outpost_arn                        = (known after apply)
+ password_data                      = (known after apply)
+ placement_group                    = (known after apply)
+ placement_group_id                 = (known after apply)
+ placement_partition_number         = (known after apply)
+ primary_network_interface_id       = (known after apply)
+ private_dns                        = (known after apply)
+ private_ip                         = (known after apply)
+ public_dns                         = (known after apply)
+ public_ip                          = (known after apply)
+ region                             = "us-west-2"
+ secondary_private_ips              = (known after apply)
+ security_groups                    = (known after apply)
+ source_dest_check                  = true
+ spot_instance_request_id           = (known after apply)
+ subnet_id                          = "subnet-0c152b773c47af446"
+ tags                               = {
```

```
    + "Environment"        = "validation"
    + "Name"               = "bloodhound-teardown-test"
    + "bloodhound:test"     = "true"
    + "bloodhound:test_type" = "teardown_validation"
  }
+ tags_all                        = {
    + "Environment"        = "validation"
    + "Name"               = "bloodhound-teardown-test"
    + "bloodhound:test"     = "true"
    + "bloodhound:test_type" = "teardown_validation"
  }
+ tenancy                         = (known after apply)
+ user_data_base64                = (known after apply)
+ user_data_replace_on_change       = false
+ vpc_security_group_ids          = (known after apply)

+ capacity_reservation_specification (known after apply)

+ cpu_options (known after apply)

+ ebs_block_device (known after apply)
```

```
      + enclave_options (known after apply)

      + ephemeral_block_device (known after apply)

      + instance_market_options (known after apply)

      + maintenance_options (known after apply)

      + metadata_options (known after apply)

      + network_interface (known after apply)

      + primary_network_interface (known after apply)

      + private_dns_name_options (known after apply)

      + root_block_device (known after apply)
    }

Plan: 1 to add, 0 to change, 0 to destroy.
```

Changes to Outputs:

+ bloodhound_test_instance_id = (known after apply)

aws_instance.bloodhound_teardown_test[0]: Creating...

aws_instance.bloodhound_teardown_test[0]: Still creating... [00m10s elapsed]

aws_instance.bloodhound_teardown_test[0]: Creation complete after 14s [id=i-01fee506f85a03005]

Apply complete! Resources: 1 added, 0 changed, 0 destroyed.

Outputs:

bloodhound_lambda_url = "https://kn5244cyenar6pzgexnrq5oh2m0ohyvb.lambda-url.us-west-2.on.aws/"

bloodhound_test_instance_id = "i-01fee506f85a03005"

lambda_function_name = "BloodhoundLambdaV2"

Capturing instance ID from Terraform output...

Instance created:

i-01fee506f85a03005

Waiting for instance to become visible to AWS APIs...

----------------------------------------

Scan Confirmation Step

----------------------------------------

Skipping Slack scan confirmation (automated validation mode).

----------------------------------------

Triggering Bloodhound Validation Teardown

----------------------------------------

Invoking Lambda validation teardown...

{
    "StatusCode": 200,
    "ExecutedVersion": "$LATEST"
}

Lambda response:

{"ok": true, "regions": ["us-east-1", "us-east-2", "us-west-1", "us-west-2"], "scan": {"candidates_total": 62, "kept_total": 1}, "budget": {"projected_month_end_spend_usd": 3677.6966075729483, "dynamic_monthly_allowance_usd": 0.0, "over_budget_threshold_met": true}, "teardown": {"apply_changes": true, "simulate": false, "targets_filter": ["i-01fee506f85a03005"], "planned_actions": 62, "execution": {"attempted": 1, "succeeded": 1, "failed": 0, "simulated": 0, "failures": [], "targets_filter": ["i-01fee506f85a03005"], "planned_actions_total": 62, "executed_actions_total": 1, "allow_all_targets": false}}}

**Validating Lambda teardown execution metrics...**

**Execution metrics:**

**attempted: 1**

**succeeded: 1**

**failed: 0**

**Lambda execution metrics validated successfully.**

**Step 6: Verifying instance deletion...**

**SUCCESS: Instance terminated confirmed.**

**RESULT: PASS**

**Bloodhound successfully deleted the resource.**

terraform_data.build_lambda_pkg: Refreshing state... [id=5ad4c1d0-1e07-ae07-cca2-ea66b293a381]

data.archive_file.lambda_zip: Reading...

data.archive_file.lambda_zip: Read complete after 0s [id=b1a59953cc20d5497c31342a65ca9eb3aaa66168]

data.aws_iam_policy_document.bloodhound: Reading...

data.aws_subnets.default: Reading...

data.aws_iam_policy_document.lambda_assume: Reading...

data.aws_ami.amazon_linux_2: Reading...

data.aws_caller_identity.current: Reading...

data.aws_iam_policy_document.lambda_assume: Read complete after 0s [id=2690255455]

data.aws_iam_policy_document.bloodhound: Read complete after 0s [id=3536511632]

aws_iam_policy.bloodhound_policy: Refreshing state... [id=arn:aws:iam::388691194728:policy/bloodhound-v2-policy]

aws_iam_role.lambda_role: Refreshing state... [id=bloodhound-v2-role]

data.aws_caller_identity.current: Read complete after 0s [id=388691194728]

terraform_data.account_guard: Refreshing state... [id=21ae472e-35bd-81b6-76c9-6a7f8b2cc399]

**aws_iam_role_policy_attachment.basic_logs: Refreshing state... [id=bloodhound-v2-role/arn:aws:iam::aws:policy/service-role/AWSLambdaBasicExecutionRole]**

**aws_iam_role_policy_attachment.bloodhound_attach: Refreshing state... [id=bloodhound-v2-role/arn:aws:iam::388691194728:policy/bloodhound-v2-policy]**

**aws_lambda_function.bloodhound_v2: Refreshing state... [id=BloodhoundLambdaV2]**

**data.aws_subnets.default: Read complete after 0s [id=us-west-2]**

**data.aws_ami.amazon_linux_2: Read complete after 1s [id=ami-0534a0fd33c655746]**

**aws_instance.bloodhound_teardown_test[0]: Refreshing state... [id=i-01fee506f85a03005]**

**aws_lambda_function_url.bloodhound_url: Refreshing state... [id=BloodhoundLambdaV2]**

**aws_lambda_permission.function_url_invoke: Refreshing state... [id=AllowFunctionInvoke]**

**aws_lambda_alias.bloodhound_prod: Refreshing state... [id=arn:aws:lambda:us-west-2:388691194728:function:BloodhoundLambdaV2:prod]**

**aws_lambda_permission.function_url_public: Refreshing state... [id=AllowFunctionUrlPublic]**

Terraform used the selected providers to generate the following execution

plan. Resource actions are indicated with the following symbols:

- destroy

Terraform will perform the following actions:

```
  # aws_instance.bloodhound_teardown_test[0] will be destroyed
  # (because index [0] is out of range for count)
  - resource "aws_instance" "bloodhound_teardown_test" {
      - ami                                  = "ami-0534a0fd33c655746" -> null
      - arn                                  = "arn:aws:ec2:us-west-2:388691194728:instance/i-01fee506f85a03005" -> null
      - associate_public_ip_address          = true -> null
      - availability_zone                    = "us-west-2a" -> null
      - disable_api_stop                     = false -> null
      - disable_api_termination              = false -> null
      - ebs_optimized                        = false -> null
      - force_destroy                        = false -> null
      - get_password_data                    = false -> null
      - hibernation                          = false -> null
      - id                                   = "i-01fee506f85a03005" -> null
```

- instance_initiated_shutdown_behavior = "stop" -> null

- instance_state = "shutting-down" -> null

- instance_type = "t3.micro" -> null

- ipv6_address_count = 0 -> null

- ipv6_addresses = [] -> null

- monitoring = false -> null

- placement_partition_number = 0 -> null

- primary_network_interface_id = "eni-05fcf317049d1646e" -> null

- private_dns = "ip-10-0-1-172.us-west-2.compute.internal" -> null

- private_ip = "10.0.1.172" -> null

- public_dns = "ec2-54-185-170-144.us-west-2.compute.amazonaws.com" -> null

- public_ip = "54.185.170.144" -> null

- region = "us-west-2" -> null

- secondary_private_ips = [] -> null

- security_groups = [] -> null

- source_dest_check = true -> null

- subnet_id = "subnet-0c152b773c47af446" -> null

- tags = {

  - "Environment" = "validation"

```
    - "Name"               = "bloodhound-teardown-test"
    - "bloodhound:test"      = "true"
    - "bloodhound:test_type" = "teardown_validation"
  } -> null
- tags_all                         = {
    - "Environment"         = "validation"
    - "Name"               = "bloodhound-teardown-test"
    - "bloodhound:test"      = "true"
    - "bloodhound:test_type" = "teardown_validation"
  } -> null
- tenancy                          = "default" -> null
- user_data_replace_on_change          = false -> null
- vpc_security_group_ids             = [
    - "sg-02161313beaae321e",
  ] -> null
  # (9 unchanged attributes hidden)

- capacity_reservation_specification {
    - capacity_reservation_preference = "open" -> null
  }
```

```
- cpu_options {
    - core_count       = 1 -> null
    - threads_per_core = 2 -> null
      # (1 unchanged attribute hidden)
  }

- credit_specification {
    - cpu_credits = "unlimited" -> null
  }

- enclave_options {
    - enabled = false -> null
  }

- maintenance_options {
    - auto_recovery = "default" -> null
  }

- metadata_options {
    - http_endpoint               = "enabled" -> null
    - http_protocol_ipv6          = "disabled" -> null
```

```
    - http_put_response_hop_limit = 1 -> null
    - http_tokens                 = "optional" -> null
    - instance_metadata_tags      = "disabled" -> null
  }

- primary_network_interface {
    - delete_on_termination = true -> null
    - network_interface_id  = "eni-05fcf317049d1646e" -> null
  }

- private_dns_name_options {
    - enable_resource_name_dns_a_record    = false -> null
    - enable_resource_name_dns_aaaa_record = false -> null
    - hostname_type                        = "ip-name" -> null
  }

- root_block_device {
    - delete_on_termination = true -> null
    - device_name           = "/dev/xvda" -> null
    - encrypted             = false -> null
    - iops                  = 100 -> null
```

```
      - tags                 = {} -> null
      - tags_all             = {} -> null
      - throughput           = 0 -> null
      - volume_id            = "vol-01e079b6b5800eff1" -> null
      - volume_size          = 8 -> null
      - volume_type          = "gp2" -> null
        # (1 unchanged attribute hidden)
    }
  }

Plan: 0 to add, 0 to change, 1 to destroy.

Changes to Outputs:
  - bloodhound_test_instance_id = "i-01fee506f85a03005" -> null
aws_instance.bloodhound_teardown_test[0]: Destroying... [id=i-01fee506f85a03005]
aws_instance.bloodhound_teardown_test[0]: Still destroying... [id=i-01fee506f85a03005, 00m10s elapsed]
aws_instance.bloodhound_teardown_test[0]: Destruction complete after 14s

Apply complete! Resources: 0 added, 0 changed, 1 destroyed.
```

**Outputs:**

**bloodhound_lambda_url = "https://kn5244cyenar6pzgexnrq5oh2m0ohyvb.lambda-url.us-west-2.on.aws/"**

**lambda_function_name = "BloodhoundLambdaV2"**

**Step 8: Reconciling Terraform state**

**terraform_data.build_lambda_pkg: Refreshing state... [id=5ad4c1d0-1e07-ae07-cca2-ea66b293a381]**

**data.archive_file.lambda_zip: Reading...**

**data.archive_file.lambda_zip: Read complete after 0s [id=b1a59953cc20d5497c31342a65ca9eb3aaa66168]**

**data.aws_subnets.default: Reading...**

**data.aws_ami.amazon_linux_2: Reading...**

**data.aws_iam_policy_document.bloodhound: Reading...**

**data.aws_iam_policy_document.lambda_assume: Reading...**

**data.aws_caller_identity.current: Reading...**

**data.aws_iam_policy_document.lambda_assume: Read complete after 0s [id=2690255455]**

**data.aws_iam_policy_document.bloodhound: Read complete after 0s [id=3536511632]**

**aws_iam_policy.bloodhound_policy: Refreshing state... [id=arn:aws:iam::388691194728:policy/bloodhound-v2-policy]**

**aws_iam_role.lambda_role: Refreshing state... [id=bloodhound-v2-role]**

**data.aws_caller_identity.current: Read complete after 0s [id=388691194728]**

**terraform_data.account_guard: Refreshing state... [id=21ae472e-35bd-81b6-76c9-6a7f8b2cc399]**

**aws_iam_role_policy_attachment.basic_logs: Refreshing state... [id=bloodhound-v2-role/arn:aws:iam::aws:policy/service-role/AWSLambdaBasicExecutionRole]**

**aws_iam_role_policy_attachment.bloodhound_attach: Refreshing state... [id=bloodhound-v2-role/arn:aws:iam::388691194728:policy/bloodhound-v2-policy]**

**aws_lambda_function.bloodhound_v2: Refreshing state... [id=BloodhoundLambdaV2]**

**data.aws_ami.amazon_linux_2: Read complete after 1s [id=ami-0534a0fd33c655746]**

**data.aws_subnets.default: Read complete after 1s [id=us-west-2]**

**aws_lambda_function_url.bloodhound_url: Refreshing state... [id=BloodhoundLambdaV2]**

**aws_lambda_alias.bloodhound_prod: Refreshing state... [id=arn:aws:lambda:us-west-2:388691194728:function:BloodhoundLambdaV2:prod]**

**aws_lambda_permission.function_url_invoke: Refreshing state... [id=AllowFunctionInvoke]**

**aws_lambda_permission.function_url_public: Refreshing state... [id=AllowFunctionUrlPublic]**

**No changes. Your infrastructure matches the configuration.**

**Terraform has compared your real infrastructure against your configuration**

**and found no differences, so no changes are needed.**

**Apply complete! Resources: 0 added, 0 changed, 0 destroyed.**

**Outputs:**

**bloodhound_lambda_url = "https://kn5244cyenar6pzgexnrq5oh2m0ohyvb.lambda-url.us-west-2.on.aws/"**

**lambda_function_name = "BloodhoundLambdaV2"**

```
----------------------------------------
Validation Complete
----------------------------------------

Teardown pipeline verified.

Keeping only the 3 most recent validation logs.

==================================================
Validation workflow completed successfully.
==================================================

Keeping only the 3 most recent workflow logs.
(myvenv) ➜  Bloodhound git:(mmc/bloodhound_v2) ✗
```